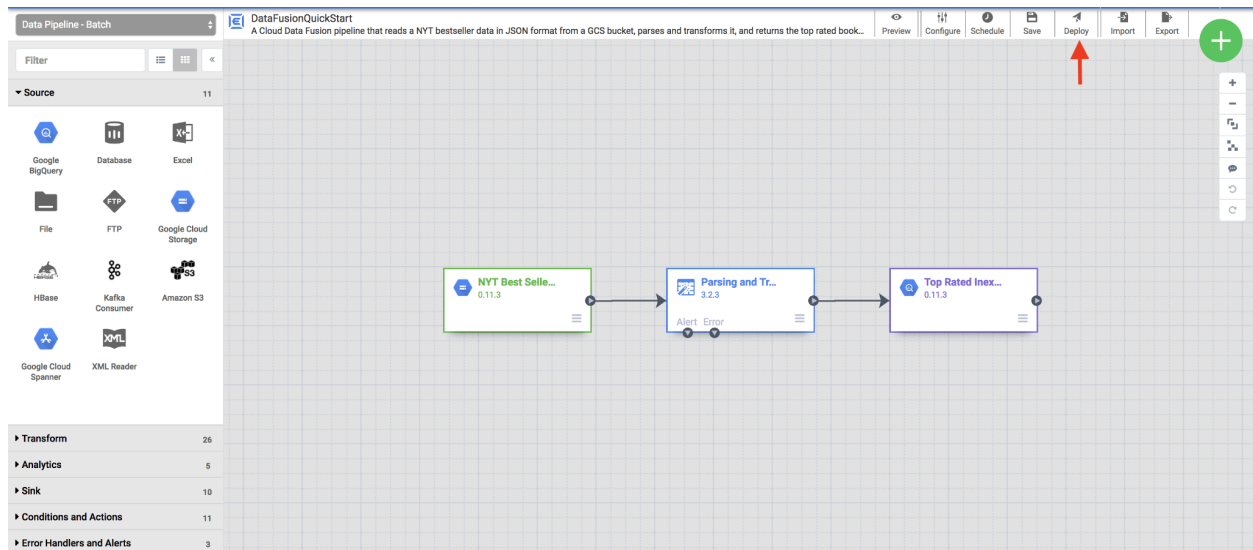
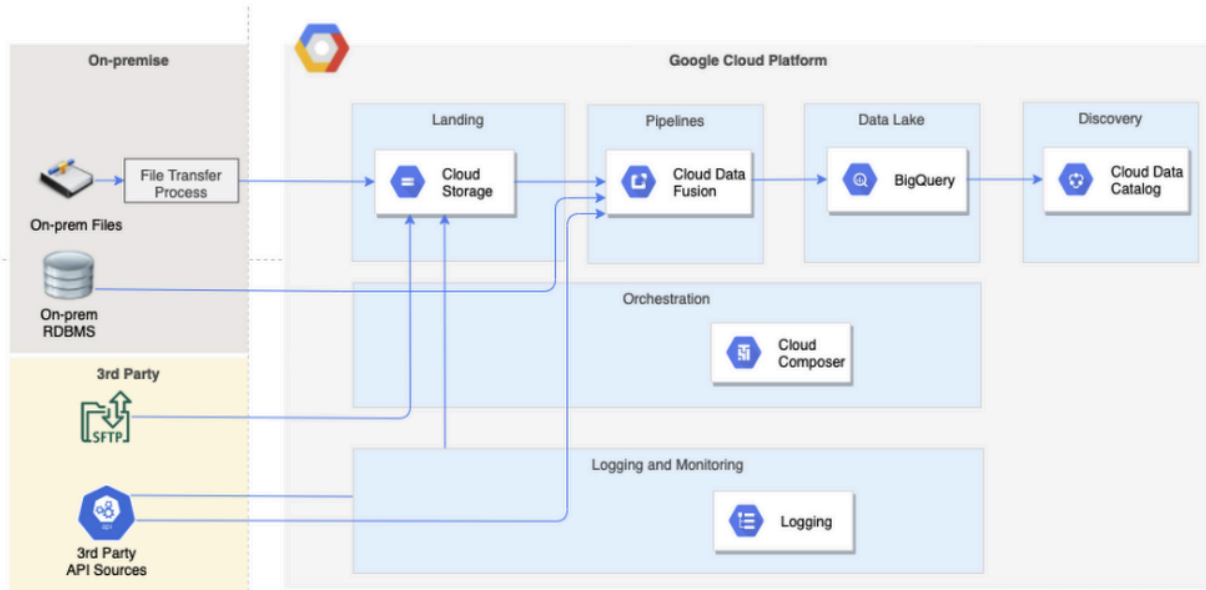
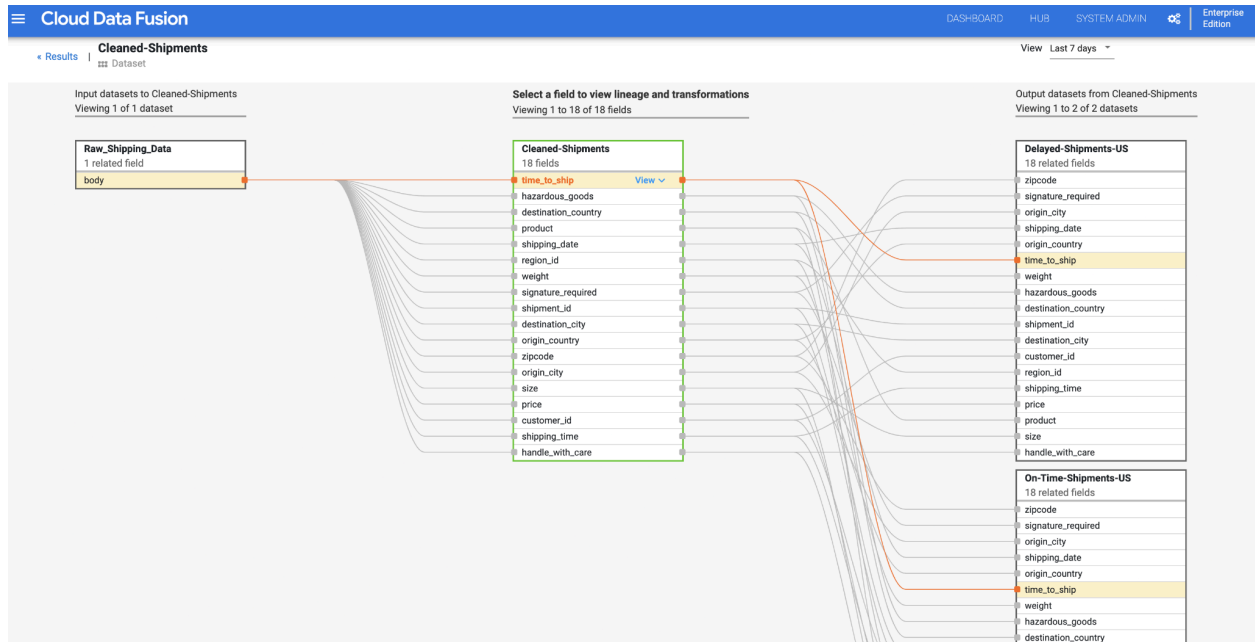


Cloud Data Fusion (CDF) – Capabilities, Architecture & Integrations





◆ Part 1: CDF in the Context of Conversational & AI Pipelines

CDF is **not** the AI engine itself, but it plays a **critical supporting role** for conversational AI systems.

◆ Response Generation (Indirect)

- CDF prepares **clean, structured datasets** used by:
 - Chatbots
 - NLP models
 - Analytics dashboards
- Feeds processed data into **BigQuery / ML pipelines**

📌 Example:

Chat logs → CDF → BigQuery → Model training → Better responses

◆ Sentiment Analysis (Data Enablement)

- CDF ingests:
 - Chat transcripts
 - Call logs
 - Feedback text
- Prepares data for:
 - Sentiment scoring
 - NLP enrichment (via ML services)

📌 Actual sentiment scoring happens in:

- Vertex AI
 - Cloud Natural Language API
- CDF handles **ETL before & after**.

♦ Speech Integration (Pipeline Level)

- CDF pipelines process:
 - Speech-to-text outputs
 - Voice transcripts
- Moves data between:
 - Storage → Analytics → ML systems

📌 Used heavily in **call center analytics**.

♦ Conversation Datasets for Model Improvement

CDF is ideal for:

- Historical conversation ingestion
- Feature extraction
- Dataset versioning
- Feeding training pipelines

📌 Enables **continuous learning loops**.

♦ Part 2: What is Cloud Data Fusion (Core Definition)

Cloud Data Fusion is a fully managed ETL/ELT service built on CDAP (Cask Data Application Platform) that enables visual, code-free data integration at scale.

◆ Part 3: Core Features Explained Clearly

1 Fully Managed ETL / ELT

- No cluster management
 - Google handles:
 - Provisioning
 - Scaling
 - Patching
-

2 Built on CDAP

- Open-source data integration platform
 - Plugin-based architecture
 - Extensible & reusable pipelines
-

3 Visual Pipeline Designer (Studio UI)

- Drag-and-drop interface
- Design pipelines visually:

Source → Transform → Sink

📌 Minimal coding, high productivity.

4 Pipeline Types

- **Batch pipelines**
 - **Real-time (streaming) pipelines**
 - **Wrangler pipelines** (data preparation)
-

5 Reusable Plugins

- Source plugins
- Transform plugins
- Sink plugins
- Custom plugins (Java)

 Promotes standardization across teams.

6 Wrangler Transformations

- Interactive data cleansing
 - Similar to Dataprep concepts
 - Ideal for:
 - Regex cleaning
 - Standardization
 - Schema fixing
-

7 Pipeline Drafts & Versions

- Save drafts
- Version pipelines
- Rollback easily

 Enterprise change management friendly.

8 Deployment Lifecycle

Draft → Test → Deploy → Monitor → Retire

9 Metadata Tracking & Lineage

- Automatic metadata capture
- Visual lineage graph:

Source → Transform → Sink

📌 Critical for governance & audits.

10 Operational Summary Dashboard

- Pipeline runs
- Failures
- Throughput
- Latency

📌 Ops teams love this.

11 Studio Interface

- Central UI for:
 - Pipelines
 - Metadata
 - Lineage
 - Monitoring

12 Namespaces

- Logical isolation:
 - Dev
 - Test
 - Prod

📌 Supports multi-team usage.

13 System vs User Pipelines

- **System pipelines:** internal operations
 - **User pipelines:** business logic
-

14 Preview Mode

- Run pipeline on **sample data**
 - Validate schema & logic
 - No full execution cost
-

15 Pipeline Templates

- Reusable pipeline blueprints
 - Faster onboarding
-

16 Execution Engines

- **Spark**
- **Dataproc**
- (Managed by CDF)

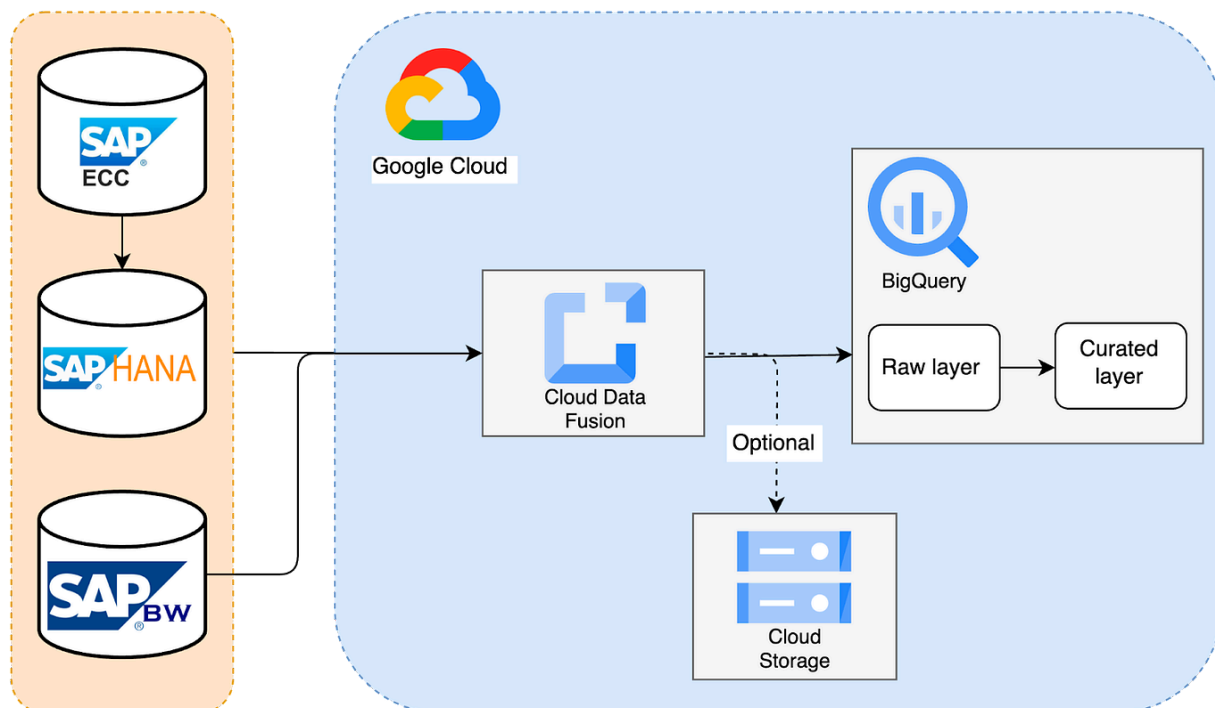
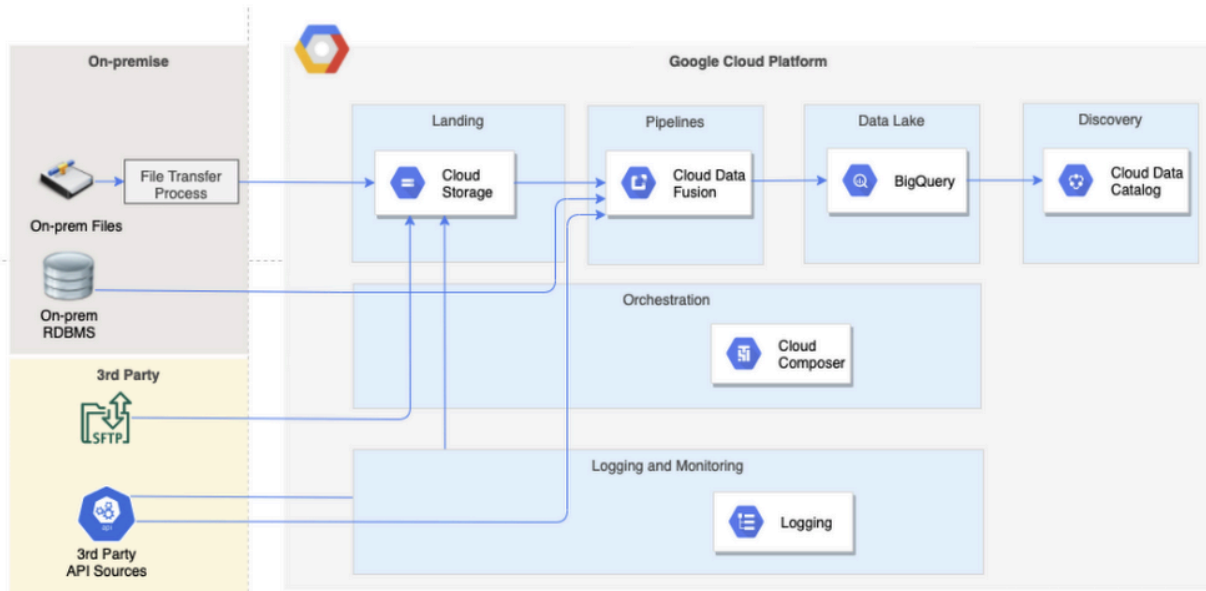
📌 Scales automatically with workload.

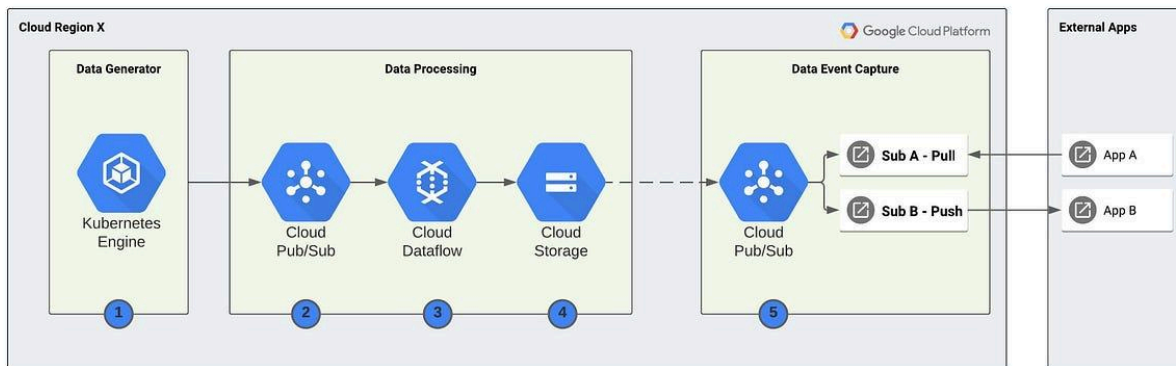
17 Scaling Clusters

- Auto-scaling Dataproc clusters
 - Cost-efficient execution
 - Handles large datasets
-

◆ Part 4: Connectivity & Integration (VERY IMPORTANT)

Built-in Connectors





4

Native GCP Connectors

- BigQuery
- Cloud Storage
- Pub/Sub
- Cloud SQL
- Dataproc
- Dataflow

External & Enterprise Connectors

- JDBC (Oracle, MySQL, Postgres, SQL Server)
- REST APIs
- SFTP
- FTP
- SaaS sources

 Makes CDF a **true integration hub**.

End-to-End CDF Architecture (Text Diagram)

Source Systems

(DB / Files / APIs / PubSub)



Cloud Data Fusion Studio
(Visual Pipelines + Wrangler)



Spark / Dataproc Execution



BigQuery / GCS / Cloud SQL



Analytics / ML / Dashboards



Interview-Ready One-Liner

Cloud Data Fusion is a fully managed, CDAP-based ETL/ELT service that provides a visual pipeline designer, reusable plugins, metadata lineage, and rich connectivity to GCP and enterprise systems, making it ideal for scalable data integration and analytics pipelines.



When to Use CDF

- ✓ Low-code ETL needed
 - ✓ Many heterogeneous sources
 - ✓ Strong governance & lineage required
 - ✓ Enterprise data integration
 - ✓ Analytics & ML data preparation
-

-  Step-by-step CDF pipeline lab
-  GCP certification MCQs
-  CDF vs Dataflow vs Dataproc
-  Printable CDF cheat sheet
-  End-to-end chatbot data pipeline using CDF